

Student Name: _____

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Official Use

☐ Complete ☐ Fix and resubmit ☐ Redo

The following struct and procedure are defined.

```
const int SIZE = 5;

struct stat_double {
    double numbers [SIZE];
    double mystery;
};

void who_am_i ( stat_double &data )
{
    int i;

    data.mystery = data.numbers [0];

    for (i = 0; i < SIZE; i++) {
        if ( data.numbers [i] > data.mystery ) {
            data.mystery = data.numbers [i];
        }
    }

    return;
}
```

1. Hand execute the code below using this procedure.

```
// Initialise d.numbers with {-20.0, 3.2, 1.9, -1.5, 1.3}
// and d.mystery with 0.0'.
stat_double d = { {-20.0, 3.2, 1.9, -1.5, 1.3}, 0.0 };

who_am_i ( d );
```

You are creating a system for managing computers in an office. The following fields are needed to represent a computer: (1) an identification number; (2) a description; (3) a location.

2. Write a data structure named `computer_data` to capture these fields.

3. Write a function to read in the information for a `number_to_read` number of computers into `computer_list`.

```
void read_computers ( computer_data &computer_list [], int number_to_read )
```

4. Write the main function which asks the user for `computer_num` number of computers. Then call your `read_computers` function to read in data for `computer_num` computers. Finally, print the list of computers.

```
int main ( )
{
    const int      MAX = 100;      // Maximum size of array of computers
    computer_data computers [MAX];  // Array of computers
    int            computer_num;    // Actual number of computers stored

    // Ask for number of computers

    // Read in computer list

    // Print list of computers
```

```
}
```